JIGME NAMGAY

Survey Engineer / Geoinformatics Engineer

Nationality: Bhutanese
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Location: Thimphu, Bhutan

PROFESSIONAL PROFILE

Survey Engineer / Geoinformatics Engineer with over three years of professional experience in geospatial engineering, Geographic Information Systems (GIS), remote sensing, spatial data management, and digital geospatial infrastructure development.

Currently serving at the National Land Commission Secretariat, Department of Survey and Mapping, contributing to national-level geospatial initiatives including Bhutan’s National Spatial Data Infrastructure (NSDI), National Land Use Zoning (NLUZ), National Land Use and Land Cover (LULC) mapping, and the Gelephu Mindfulness City (GMC) initiative.

Experienced in GIS-based spatial analysis, remote sensing applications, satellite image processing, geospatial database management, web GIS development, cartographic production, and spatial data quality assessment. Skilled in ArcGIS Pro, QGIS, ArcGIS Online, Google Earth Engine, Python-based GIS workflows, and modern web technologies.

Committed to advancing geospatial solutions through Earth Observation, spatial data science, automation, artificial intelligence applications, and digital transformation for sustainable land management and decision-making.

PROFESSIONAL EXPERIENCE

National Land Commission Secretariat (NLCS)

Department of Survey and Mapping

Thimphu, Bhutan

Survey Engineer / Geoinformatics Engineer

Geo-informatics Division

01 January 2024 – Present

Key Responsibilities

- Support national geospatial projects through GIS analysis, remote sensing, spatial data processing, and geospatial information management.
- Perform spatial data quality control including topology correction, attribute validation, and dataset verification using ArcGIS Pro and QGIS.
- Process satellite imagery and support Land Use and Land Cover (LULC) mapping activities.
- Develop thematic maps, spatial visualizations, and cartographic outputs for planning and decision support.
- Support geospatial database management and preparation of spatial datasets according to national standards.
- Assist in the development of web-based GIS applications and geospatial visualization platforms.
- Apply Python-based automation techniques to improve GIS workflow efficiency.
- Collaborate with government agencies and technical teams in implementing geospatial solutions.

KEY PROJECT EXPERIENCE

Bhutan National Spatial Data Infrastructure (NSDI)

- Contributed to the development and enhancement of Bhutan’s National Spatial Data Infrastructure platform.
- Supported integration, management, and visualization of national geospatial datasets.
- Developed frontend components for geospatial applications using modern web technologies.
- Applied Git and GitHub workflows for collaborative development and version control.
- Supported improved accessibility and visualization of spatial information.

Gelephu Mindfulness City (GMC) Project

- Developed an interactive Web GIS application to support visualization and communication of geospatial information.
- Integrated spatial datasets from multiple sources for project visualization and analysis.
- Supported GIS data preparation and presentation of project outputs during the Bhutan Innovation Forum.

National Land Use Zoning (NLUZ)

- Supported national land-use planning activities through GIS-based spatial analysis.
- Applied Multi-Criteria Decision Analysis (MCDA) approaches for identification and evaluation of alienable land.
- Conducted spatial data preparation, verification, and analysis to support land management decisions.

National Land Use and Land Cover (LULC) Mapping

- Supported national LULC mapping initiatives through satellite image analysis and GIS workflows.
- Processed Sentinel-2 satellite imagery for land cover assessment.
- Conducted spatial data validation, topology correction, and quality assurance.
- Supported preparation and maintenance of national land cover datasets.

Survey Engineer

Topographical Survey Division

Department of Survey and Mapping
National Land Commission Secretariat

01 January 2023 – 01 January 2024

- Managed and monitored Continuous Operating Reference System (CORS) infrastructure supporting national surveying activities.
- Supported geospatial data preparation, maintenance, and validation for mapping activities.
- Assisted in integrating survey-related datasets into GIS environments.
- Supported technical workflows related to surveying, mapping, and spatial data management.

GIS Intern

Ministry of Infrastructure and Transport

(formerly Ministry of Works and Human Settlement)
Thimphu, Bhutan

01 May 2020 – 15 June 2020

- Supported GIS-based urban expansion boundary demarcation activities.
- Assisted in spatial data preparation, digitization, editing, and map production.
- Supported development of GIS databases using ArcGIS.
- Contributed to the Street Addressing System project for Phuentsholing through GIS database preparation and spatial data organization.

GIS Intern

Ministry of Works and Human Settlement

Thimphu, Bhutan

June 2019 – July 2019

- Supported GIS data collection and preparation for urban mapping projects.
- Assisted in ArcGIS geodatabase development and spatial data management.
- Performed GIS editing, digitization, and quality control activities.

EDUCATION

Bachelor of Technology (Geo Informatics Engineering)

University of Petroleum and Energy Studies (UPES)
Dehradun, India

03 June 2017 – 03 June 2021

Scholarship: Royal Government of Bhutan Scholarship Recipient

Major Areas of Study

- Geographic Information Systems (GIS)
- Remote Sensing and Satellite Image Processing
- Digital Photogrammetry
- Spatial Database Management
- Coordinate Reference Systems and Geospatial Projections
- Digital Image Analysis
- Geostatistical Analysis
- Cartography

PROFESSIONAL CERTIFICATIONS AND CONTINUOUS LEARNING

Advanced Remote Sensing

Spatial Analysis using Artificial Intelligence, Machine Learning and Deep Learning

Issued by:
Indian Institute of Remote Sensing (IIRS),
Indian Space Research Organisation (ISRO),
Department of Space, Government of India

Training Duration: 2 Weeks
03 March 2025 – 14 March 2025
Dehradun, India

Capacity Development on Fit-for-Purpose Land Use for National Land Use Zoning (NLUZ)

Issued by:
Asian Institute of Technology (AIT),
School of Engineering and Technology, Thailand

Training Duration:
07 May 2025 – 14 May 2025

Focus Areas:

- Fit-for-purpose land use planning
- GIS applications for land management
- National Land Use Zoning methodologies

Decentralized Application Development

Issued by:
Gyalpozhing College of Information Technology (GCIT)

Training Duration:
21 April 2025 – 24 April 2025

Programming for Everybody (Getting Started with Python)

Issued by:
Coursera | University of Michigan

Completed:
23 June 2020

Focus Areas:

- Python programming fundamentals
- Variables, functions, loops, and data structures
- Problem-solving using programming concepts

Fundamentals of GIS

Issued by:
Coursera | University of California, Davis

Completed:
19 June 2020

Focus Areas:

- GIS concepts and applications
- Spatial data models
- Coordinate systems
- Basic spatial analysis

Introduction to GIS Mapping

GIS, Mapping, and Spatial Analysis Specialization

Issued by:
Coursera | University of Toronto

Completed:
26 May 2020

Focus Areas:

- GIS mapping principles
- Spatial visualization
- Geographic data interpretation

Spatial Data Science: The New Frontier in Analytics

Professional Online Course

Duration: 6 Weeks
Completed: 12 October 2023

Focus Areas:

- Spatial analytics
- Data-driven geographic analysis
- Emerging spatial data science approaches

Basics of JavaScript Web Apps

Online Web Development Course

Completed:
05 May 2024

Focus Areas:

- JavaScript fundamentals
- Web application concepts
- Client-side programming basics

Creating Geospatial and Location Intelligence Applications and Handling of Big Data

Issued by:
Hexagon in partnership with University of Petroleum and Energy Studies (UPES)

Industrial Program Attendance:
17 June 2021 – 18 June 2021

Focus Areas:

- Geospatial applications
- Location intelligence
- Big data handling in geospatial environments

TECHNICAL SKILLS

GIS and Spatial Technologies

- ArcGIS Pro
- ArcGIS Online
- QGIS
- ERDAS Imagine
- TerrSet
- Google Earth Engine
- Google Earth Pro
- Spatial Analysis
- Terrain Analysis
- Cartography
- Web GIS
- Geospatial Data Management

Remote Sensing and Earth Observation

- Satellite Image Processing
- Sentinel-2 Data Analysis
- Land Use and Land Cover Mapping
- Digital Image Analysis
- Remote Sensing Applications
- AI/ML Applications in Remote Sensing

Programming and Digital Technologies

- Python
- JavaScript Fundamentals
- GIS Workflow Automation
- Astro Framework
- Git and GitHub
- Web Application Development

Data Management

- Spatial Database Concepts
- Geospatial Data Quality Control
- Metadata Management
- Vector Data Processing
- Attribute Validation

CORE COMPETENCIES

- Geospatial Engineering
- GIS Analysis
- Remote Sensing Applications
- Land Use Planning
- Spatial Data Science
- Web GIS Development
- Geospatial Database Management
- Cartographic Production
- Technical Documentation
- Research and Analytical Thinking
- Government Project Implementation
- Cross-disciplinary Collaboration

LANGUAGES

Dzongkha: Native
English: Professional Working Proficiency (EF SET C1 Advanced)
Hindi: Working Proficiency

REFERENCES

Available upon request